A safety aspect, which has not been thoroughly studied, is the rapidly changing strong magnetic field (1 T s^{-1}), which will put great stress on the coils and structural material. The effects of such high fields on the operators are unknown.

19.18. NUCLEAR EXPLOSIVES

Assume that 50 kg of metallic ^{235}U is rapidly brought together to provide a critical configuration. Using a generation time of 3×10^{-9} s for neutrons, it can be calculated with eqns. (19.33) and (19.34) that it would take 0.2×10^{-6} s to increase the number of neutrons from one to that required for fissioning all ^{235}U atoms. However, long before this time, the energy absorbed in the material would have blown it apart, so 100% efficiency is impossible in a nuclear explosion. If 2% of the 50 kg of uranium had been fissioned, the amount of energy released corresponds to about 20 000 t of TNT (1 t of TNT corresponds to 4.19×10^9 J). The energy production is said to be 20 kt. Only ^{233}U, ^{235}U, and ^{239}Pu (of the more easily available long-lived nuclides) have sufficiently high cross sections for fission by fast neutrons to have a reasonable critical mass for use as nuclear explosives. However, it is a common misconception that when a critical mass is reached, a nuclear explosion will follow. In the cases when a critical mass has been reached by mistake, no energetic nuclear explosion has ever occurred.

A rough estimate of the critical radius of a homogeneous un-reflected reactor may be obtained simply by estimating the neutron mean free path according to (11.6). Assuming ^{235}U metal with a density of 19 g cm^{-3} and a fast fission cross section of 2×10^{-24} cm^2, one obtains $\Sigma_f^{-1} = 10$ cm. A sphere with this radius weighs 80 kg. For an un-reflected metal sphere containing 93.5% ^{235}U the correct value is 52 kg. ^{239}Pu has the smallest un-reflected critical size; for ^{239}Pu (δ-phase, density 15.8 g cm^{-3}) it is 15.7 kg (~6 kg reflected), and for ^{233}U 16.2 kg (~6 kg reflected).

Assume two half-spheres of metallic ^{235}U, each of which is less than critical size but which together exceed criticality. As the distance between the two spheres is diminished the k_{eff} increases for the total system. At a certain distance k_{eff} can be maintained close to 1 but the chain reaction still be prevented. If the distance between the two half-spheres is diminished rapidly to less than this critical distance in the absense of any neutron, a single neutron can then trigger a rapidly multiplying chain reaction. Since stray neutrons are always available, e.g. from spontaneous fission, a chain reaction would be started when the distance between the spheres is less than the critical distance. Hence it is important to achieve the supercritical configuration fast compared to the average time interval between the stray neutrons.

In order to achieve a substantial energy release in the short time available before the supercritical configuration is blown apart, an external or internal neutron source is triggered. By this action the fission chain reaction starts with not a few, but with very many high energy neutrons. This greatly increases the number of neutrons generated in

the time available, causes many more fissions, and yields a correspondingly larger energy release.

Nuclear fission weapons contain "normally" either highly enriched ^{235}U (>93%) or fairly pure ^{239}Pu (≥95%). The Pu-metal has to be stabilized by alloying in order to prevent any phase changes that world convert it to powder. Some early nuclear explosives operated by having two subcritical parts of a sphere blown together by the use of chemical explosives (*gun type*), see Figure 19.19. To increase the explosive yield the fissile material is often surrounded by a heavy material (*tamper*) which has the double function of reflecting the neutrons and of acting as a heavy mass (inertia) to increase the time in which the supercritical configuration is held together. If natural uranium is used as the tamper, secondary fission in this shell can contribute to the overall energy release. However, only a fraction of the fission neutrons have energies large enough to fission ^{238}U efficiently.

A critical mass of material can also be arranged as a subcritical spherical shell which can be compressed into a supercritical sphere. This process is called *implosion* and is probably used in most weapons today, see Figure 19.19. The bomb over Hiroshima

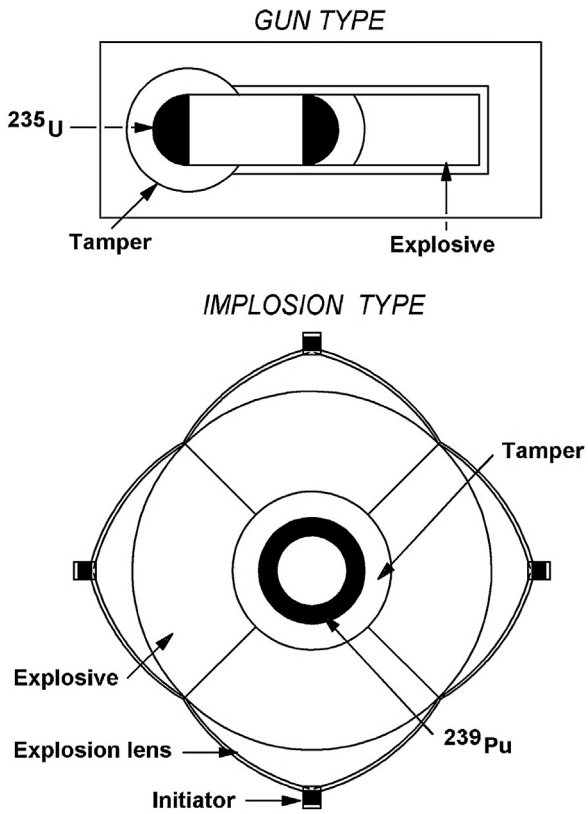


Figure 19.19 Two principles of fission bombs.

(~15 kt) used uranium and was of the gun type, while the Nagasaki bomb (~22 kt) was of the more efficient implosion type using plutonium. At the end of the implosion the density of the fission charge (U or Pu) may be increased by compression thus reducing the amount needed.

In order for these devices to function efficiently, the neutron multiplication must not begin until the critical size has been well exceeded. Furthermore, the number of neutrons in the initial generation must be sufficiently high in order to reach a very high n-flux in the short lifetime of the supercritical configuration. Therefore, as mentioned above, a few microseconds before maximum criticality a large number of fast neutrons is injected by a special neutron source initiating a rapid neutron multiplication.

The amount of ^{240}Pu in a plutonium charge must be maintained relatively low since this nuclide decays partly through spontaneous fission with the resultant release of neutrons. If the concentration of ^{240}Pu is too high, the neutron multiplication will start too early and with too few neutrons in the first generation, and the energy release is decreased. Also ^{241}Pu, if present in larger percentage, could cause problems because of its rather rapid decay to ^{241}Am. To minimize the ^{240}Pu and ^{241}Pu content, plutonium for weapons used to be produced in special reactors by low burnup (<2000 $MW_{th}d\ t^{-1}$); this is uneconomical to the nuclear power industry where high burnup of the fuel (leading to higher ^{240}Pu concentrations; Fig. 19.7) is a necessity in order to obtain the lowest electricity production costs. It has been stated that also power reactor grade plutonium, even as oxide, can be used in a weapon, though less effective and predictable. The critical mass for such plutonium containing 20% ^{240}Pu has been estimated to be 20 kg for metal and 37 kg for PuO_2. If the Pu contains a measurable concentration ^{241}Pu it must be purified at regular intervals from the ^{241}Am which grows in. In the US this was mainly made at Rock Flats by extraction of the americium from the molten Pu-metal by a chloride melt.

Nuclear weapons of the fission type have been developed in sizes ranging from 0.001 kt to about 500 kt yield. Because the critical amount can never be less than several kilograms of fissile material, a weapon of low yield such as 0.001 kt has very low efficiency ($<5 \times 10^{-6}$). The highest achieved efficiency of a plutonium based charge seems to be an order of magnitude higher than that achieved by a ^{235}U based charge. However, most fission devices are assumed to contain both materials. Eight countries (USA, USSR, UK, France, China, South Africa, Pakistan, India, and North Korea) have carried out >1000 known nuclear weapons tests (fission and fusion devices). Since the Partial Test Ban Treaty in 1963, most explosions have been underground. At the time of writing there is an agreement to ban further tests and South Africa has handed over it nuclear weapons. However, it is uncertain if this has a sufficient effect on all nations aspiring to be nuclear powers.

In the explosion of fission weapons a temperature of approximately 10^8 K is obtained. This temperature is sufficient for producing fusion reactions between deuterium and

tritium (see Ch. 17). Although no official information is available on how modern fusion weapons (hydrogen bombs) are constructed, it is generally assumed that they are based on the principle of using a fission charge as the initiator of the fusion reaction. In the debris of hydrogen bombs, lithium has been discovered. It is therefore believed that in hydrogen bombs, deuterium is combined with lithium in the form of solid LiD. ^{6}Li will react with a neutron to form tritium in the reaction

$$^{6}\mathrm{Li} + \mathrm{n} \rightarrow \mathrm{T} + {}^{4}\mathrm{He} \quad Q = 4\ \mathrm{MeV} \tag{19.48}$$

With fast neutrons the reaction

$$^{7}\mathrm{Li} + \mathrm{n(fast)} \rightarrow \mathrm{T} + {}^{4}\mathrm{He} + \mathrm{n} \tag{19.49}$$

also occurs.

The deuterium and the tritium can react as shown by equations (12.13) – (12.24). If the temperature and pressure is high enough the D-D reaction can also contribute. Natural uranium, depleted uranium, or enriched uranium must probably be used to improve the neutron economy when the charge contains LiD. Secondary fission in uranium also leads to an increased energy production in the weapon. The high energy of the neutrons from the fusion reactions make them very effective for fission of ^{238}U. The fission of ^{238}U produces large amounts of radioactive fission products, resulting in a "dirty" weapon. By contrast, if a tamper of a non-fissionable heavy material is used the only fission products released are those obtained from the ignition process and the weapon is relatively "clean".

In order to explode a hydrogen bomb, a small fission device is used. The X- and γ-rays from the fission charge travels much faster than the shock wave from the explosion and also faster than the neutrons. In order to achieve a secondary implosion, the fusion charge is surrounded by styrene foam which is converted to a plasma by the impinging X- and γ-rays. This plasma heats and compresses the thermonuclear charge which is then hit by the neutrons. This principle has been described several times in the open literature and is illustrated in Figure 19.20. By repeating this structure along the bomb axis multiple fusion (two and three stage) designs have been made.

Charges of sizes up to ~60 000 kt TNT have been tested. If it is assumed that half of the energy of a 50 Mt device comes from fusion and the other half from fission, and a 25% efficiency in the explosion, about 8 t of nuclear material is required. It is believed that the 50Mt bomb was a three stage design using ^{238}U cores in the center of the fusion charges.

Very high temperatures are reached in the center of a nuclear explosion (~10^{8} K). The exploded material forms a fireball which rapidly expands and moves upward in the atmosphere. After the first wave has passed, a low pressure area is formed under the fireball which draws material from the ground to produce the typical

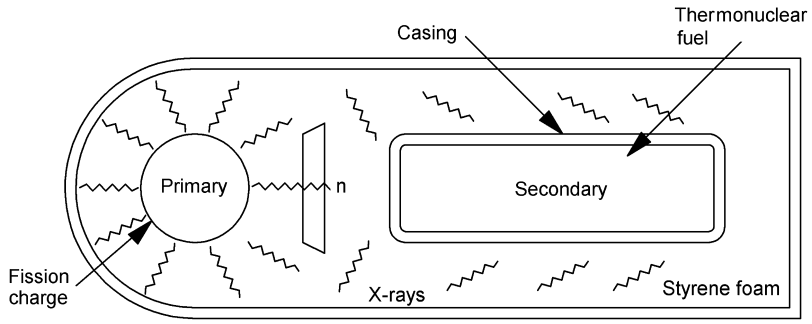


Figure 19.20 Principle of a hydrogen bomb using LiD fuel and styrene foam to produce the compressing and igniting plasma.

mushroom-like cloud associated with nuclear explosions. This cloud contains fission products as well as induced radioactivity from the construction material of the device and from the immediate surrounding ground that has been irradiated. Some of the radioactive particles are carried to great heights and eventually fall from the atmosphere as *radioactive fallout*, see Ch. 22. Much of the fallout returns to the earth close to the explosion site, but some of the material reaches the stratosphere and returns to the earth only slowly over a long period of time and over great distances from the original explosion site. The radioactivity of the fallout decays with time according to the approximate relationship

$$A = A_0 \, e^{-1.2t} \tag{19.50}$$

where A_0 is the radioactivity immediately after the explosion and t is the time in hours after the explosion.

The energy produced in the explosion of a 20 kt weapon is distributed approximately as follows: 50% pressure; 35% heat; 5% instantaneous radiation; 10% radiation from fission products. The radiation dose at 500 m from hypocenter (the vertical point on ground below the explosion center) has been estimated to be ~70 Gy; at about 1.1 km it is ~4 Gy. The pressure wave moves from the explosion site with the velocity of sound. This initial short pressure front is followed by a low pressure wave leading to a rapid change in wind direction. The deaths in Hiroshima and Nagasaki resulted in 20–30% of the cases from primary burns, 50–60% from mechanical injuries and secondary burns, and only ~15% from radiation injuries. The health detriment to the Japanese population is discussed in Chapter 15.

Table 1.1 lists some historical "firsts" with regard to nuclear weapons. The extensive tests in the atmosphere up to 1963 lead to a large global spread of tritium, ^{14}C, fission products and actinides. Scientists have used this to learn more about global wind and water currents. Radiochemists have studied the migration of deposited radionuclides, as

discussed in Chapter 22, radioecologists the uptake of radioactive elements by plants and animals, as described in Chapter 15, etc.

A number of ideas have been proposed for the peaceful use of nuclear explosives. It has been estimated that the explosive power of nuclear charges is sufficiently great that their use for large scale explosions would be much cheaper than the use of conventional explosives. About 136 explosions have been used for civil engineering projects in the former USSR; sizes varied from <1 kt to ~ 50 kt. In the United States crater formation and gas stimulation experiments have been made. Suggested uses include the blasting of harbours on remote coasts to allow greater access to inland mineral deposits, crushing of ore bodies in order to obtain valuable minerals, crushing of underground rocks which may hinder the use of natural gas reservoirs, diverting asteroids threatening the earth, etc.

19.19. EXERCISES

19.1. If the energy developed by a 20 Mt fusion weapon could be used for producing electricity at a value of 2 cents kWh^{-1} what would the "electric value" of the device be? Such a weapon may be expected to cost \$ 10 million. One ton TNT releases 1 Gcal of energy in an explosion.

19.2. The bomb over Hiroshima contained ^{235}U. How many grams were fissioned to correspond to 15 kt of TNT?

19.3. Compare two 500 MW_e electric power stations, one burning oil and the other using 3.0% enriched uranium. Both stations operate 6000 h y^{-1} at 35% efficiency (heat to electricity). The oil (43.5 MJ/kg combustion energy) is carried by 100 000 ton d.w. (dead weight, i.e. carrying capacity) oil tankers, and the uranium fuel by train cars of 20 t capacity each. (a) How many oil tankers will be needed every year for the oil-fired station? How many train cars will be needed every year for the nuclear power station for transporting (b) the enriched UO_2 reactor fuel, (c) the corresponding amount of natural uranium as U_3O_8 to the isotope enrichment plant, if the tail is 0.35% in ^{235}U? (See Ch. 3 and §19.9.) Reactor fuel rating 40 $MW_{th}d/kg$ U.

19.4. Calculate the number of collisions required to reduce a fast fission neutron ($\bar{E} =$ 2 MeV) to thermal energy (E_n 0.025 eV) in a light-water-moderated reactor, assuming that the data in Table 19.3 are valid.

19.5. Calculate the thermal fission factor for a mixture of 60% ^{239}Pu, 30% ^{240}Pu, and 10% ^{241}Pu.

19.6. The world's first water boiler reactor (LOPO, Los Alamos, 1944) was a homogeneous solution of enriched uranium sulfate as follows: 580 g ^{235}U, 3378 g ^{238}U, 534 g S, 14068 g O, and 1573 g H. From these values, and Tables 19.2 and 19.3, calculate η and f; neglect S. With $p = 0.957$, what will k_∞ be?